# Doubly Robust Estimation of Counterfactual Policy Mean Embeddings

*NeurIPS 2025*

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October 15, 2025

# Outline


1. Motivations
2. Counterfactual Policy Mean Embeddings
3. A plug-in estimator
4. An efficient influence function based estimator
5. Applications

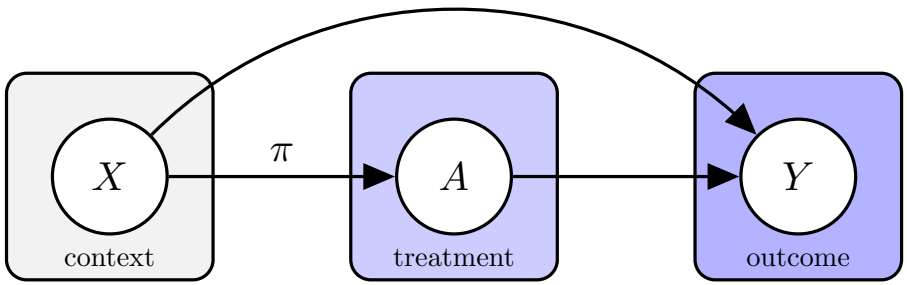
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# Setting

We are given an observational dataset $\mathcal{D}_0 = (x_i, a_i, y_i)_{i=1,\dots,n}$, for each interaction the system:

- observed a context $x_i \in \mathcal{X}$, and $x_i \sim P_X$,
- sampled an action (treatment) $a_i \in \mathcal{A}$ from a logging policy $\pi_0$ with propensity $\pi_{i,0} = \pi_0(a_i|x_i)$,
- observes a loss (outcome) $y_i \sim P_Y(\cdot|x_i, a_i)$

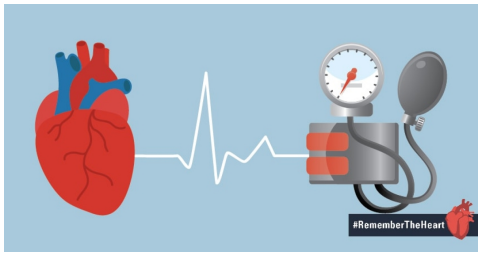



Counterfactual question: *"What would have been the outcome of the experiment, should we treat with stochastic intervention $\pi'$?"*

# Outcomes

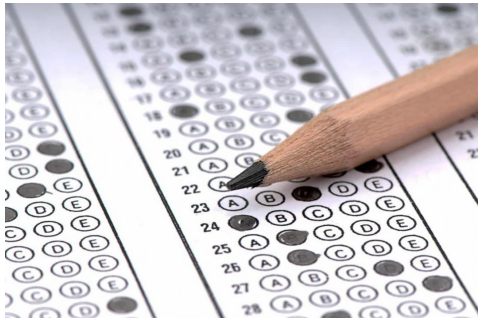
Traditionally, causal inference and policy evaluation target **scalar outcomes**:

- Health: blood pressure, cholesterol level, survival time.
- Economics: income, employment status, GDP growth.
- Education: test scores, years of schooling.









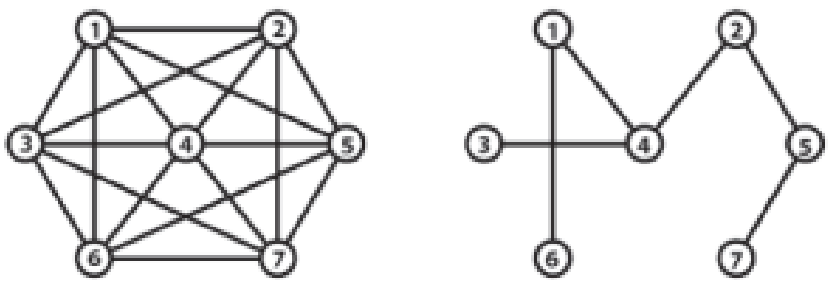
# Structured Outcomes: Medical Imaging

- Treatment/Policy: choice of treatment (choice of drug, dosage of the drug).
- Outcome: **follow-up MRI/CT scan**.
- Goal: compare counterfactual distributions of lesion shapes, sizes, textures.



# Structured Outcomes: Graphs

- Treatment/Policy: recommendation algorithm (who to follow, what to read).
- Outcome: **user's social network graph**.
- Question: do policies shift network topology (e.g., dense, sparse)?




Dense Sparse

# Structured Outcomes: Text

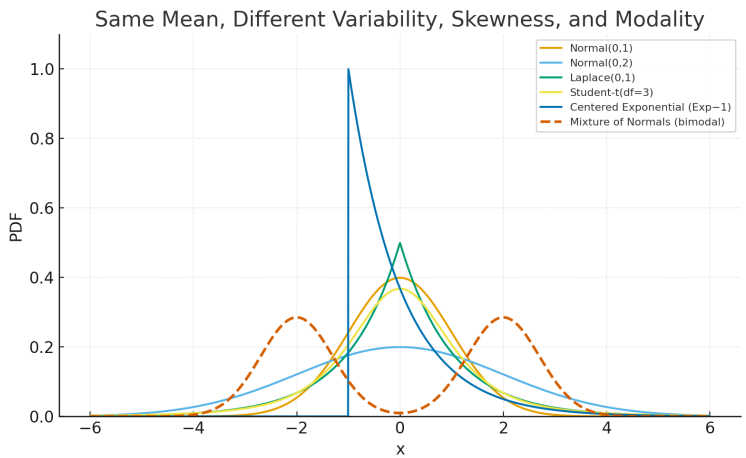
- Treatment/Policy: which message / text the system shows to the user.
- Outcome: **the user's written reply**.
- Goal: evaluate how alternative policies shift the distribution of replies (tone, length, engagement, sentiment).

**Policy $\pi$ (Neutral messages)**
Prompt: *"Would you like to hear more about healthy diets?"*
User reply: *"Maybe, I'm not sure."*

**Policy $\pi'$ (Engaging messages)**
Prompt: *"Discover how small changes in your diet can boost your energy!"*
User reply: *"Sounds interesting, tell me more!"*

# Why look beyond counterfactual expectations?

- Classical causal inference tackles the **expected value function**: $\mathbb{E}[Y(\pi)]$.
- For **scalar outcomes**, this **expectation** often suffices (e.g. mean blood pressure reduction).
- However, for **structured outcomes** (e.g. images, text, graphs), averages are meaningless:
  - There is no such thing as an "average image" or "average essay".
- Furthermore, for all type of outcomes, the **entire outcome distribution** matters:
  - A policy may not change the mean outcome distribution, but it may **reduce variability** (risk reduction).
  - A policy may alter the **shape or tails** of the distribution (e.g. rare adverse events).




- We need tools to compare the **entire counterfactual distributions**.

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# Counterfactual Policy Mean Embedding (CPME)

- Let $Y(a)$ denote the potential outcome under treatment $a$. A kernel $k_{\mathcal{Y}} : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$ defines an RKHS $\mathcal{H}_{\mathcal{Y}}$.
- **Counterfactual Policy Mean Embeddings**[1] embed the distribution of $\{Y(a)\}_{a\sim\pi}$:

$$\chi(\pi) := \mathbb{E}_{P_Y \times \pi \times P_X}[k_{\mathcal{Y}}(\cdot, Y(a))] \in \mathcal{H}_{\mathcal{Y}}.$$

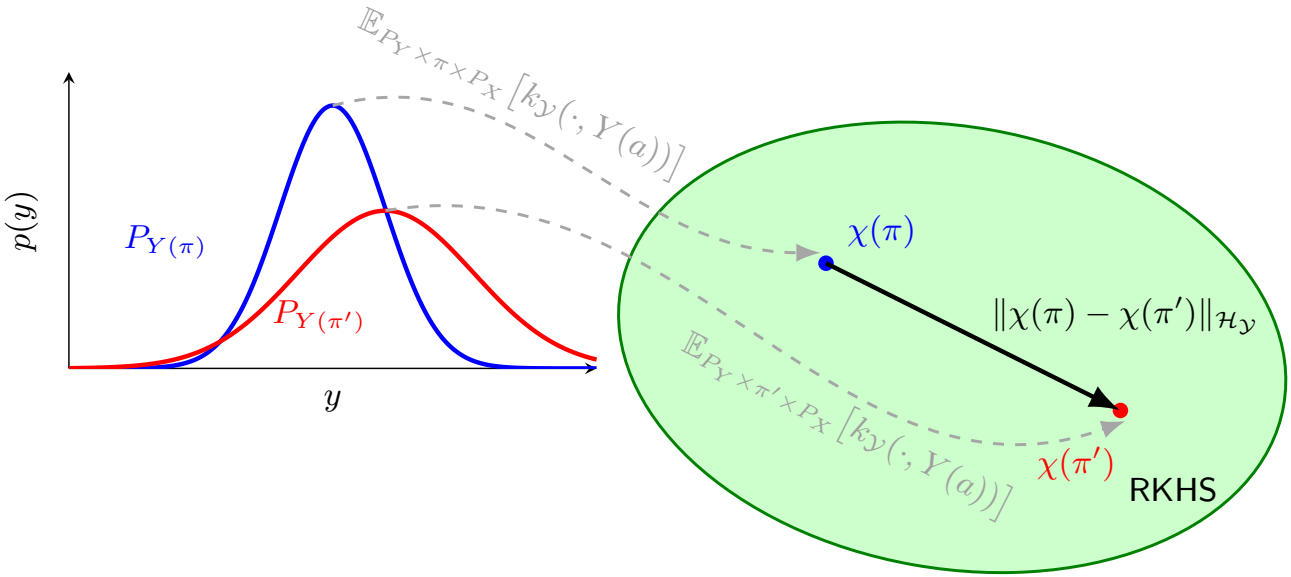
- Properties:
  - If $k$ is *characteristic*, $\chi(\pi)$ uniquely identifies the distribution of $\{Y(a)\}_{a\sim\pi}$.
  - Rich, nonparametric representation of counterfactual distributions.
- **Assumptions (selection on observables).**
  1. Consistency / no interference: $Y = Y(a)$ when $A = a$.
  2. Conditional exchangeability: $Y(a) \perp A \mid X$.
  3. Strong positivity: $\inf_{a,x} \pi_0(a \mid x) > 0$ (essential infimum under $P_X$).
- Then, under the assumptions above, the **identified** counterfactual policy mean embedding writes:

$$\begin{aligned} \chi(\pi) &= \mathbb{E}_{\pi \times P_X}\big[\mu_{Y|A,X}(a,x)\big], \\ \mu_{Y|A,X}(a,x) &:= \mathbb{E}\big[k_{\mathcal{Y}}(\cdot, Y) \mid A = a,\ X = x\big] \in \mathcal{H}_{\mathcal{Y}}, \end{aligned}$$

[1] Our setting extends Counterfactual Mean Embeddings of Muandet et al. (2021) to off-policy evaluation.

# Counterfactual Policy Mean Embedding

- **Intuition**
  - $\chi(\pi)$ is a representation of the entire distribution of $\{Y(a)\}_{a\sim\pi}$.
  - Each distribution becomes a point in the Hilbert space $\mathcal{H}_\mathcal{Y}$.
  - Distances in $\mathcal{H}_\mathcal{Y}$ correspond to differences between distributions.
  - $\|\chi(\pi) - \chi(\pi')\|_{\mathcal{H}_\mathcal{Y}}$ is zero iff distributions match (characteristic $k_\mathcal{Y}$); larger norms $\Rightarrow$ stronger distributional shift.




- **Why it is suited to structured outcomes:** choose $k_\mathcal{Y}$ to encode geometry of images/text/graphs (e.g., RBF on embeddings, string/graph kernels), then $\chi(\pi)$ summarizes the whole counterfactual distribution in that geometry.

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# A Plug-in CPME estimator

- **Regularity (RKHS).** Assume bounded, continuous kernels $k_{\mathcal{AX}}$ and $k_{\mathcal{Y}}$; measurability; and $k_{\mathcal{Y}}$ characteristic.
- Define the **Conditional mean embedding**:

$$\mu_{Y|A,X}(a,x) \;:=\; \mathbb{E}\big[k_{\mathcal{Y}}(\cdot,Y) \mid A=a,\; X=x\big] \;\in\; \mathcal{H}_{\mathcal{Y}},$$

- **Conditional mean operator (CMO)**. The CME defines a bounded linear operator

$$C_{Y|A,X} : \mathcal{H}_{\mathcal{AX}} \;\to\; \mathcal{H}_{\mathcal{Y}}, \qquad C_{Y|A,X}\,\phi_{\mathcal{AX}}(a,x) \;=\; \mu_{Y|A,X}(a,x),$$

  where $\phi_{\mathcal{AX}}(a,x) = k_{\mathcal{AX}}(\cdot,(a,x))$ is the joint feature map. Hence, for any distribution over $(A,X)$,

$$\mathbb{E}_{A,X}\left[\mu_{Y|A,X}(A,X)\right] \;=\; C_{Y|A,X}\;\mathbb{E}_{A,X}\left[\phi_{\mathcal{AX}}(A,X)\right].$$

- Then, when introducing the following kernel mean embedding:

$$\mu_{\pi} \;:=\; \mathbb{E}_X\;\mathbb{E}_{a\sim\pi(\cdot|X)}\big[\phi_{\mathcal{AX}}(a,X)\big] \;\in\; \mathcal{H}_{\mathcal{AX}},$$

- we obtain the following **decoupled formulation of CPME**

$$\boxed{\chi(\pi) \;=\; C_{Y|A,X}\,\mu_{\pi}} \in\; \mathcal{H}_{\mathcal{Y}}$$

# Plug-in Estimator: closed form solutions

**Goal.** Estimate $C_{Y|A,X}$ and $\mu_\pi$, then plug in:

$$\boxed{\hat{\chi}_{\mathrm{pi}}(\pi) \;=\; \hat{C}_{Y|A,X}\,\hat{\mu}_\pi} \;\in\; \mathcal{H}_{\mathcal{Y}}.$$

**Estimate $\hat{C}_{Y|A,X}$ (operator ridge).** For $\lambda > 0$,

$$\hat{\mathcal{L}}^c(C) \;=\; \frac{1}{n}\sum_{i=1}^{n} \left\| k_{\mathcal{Y}}(\cdot, y_i) \;-\; C\, k_{\mathcal{A}\mathcal{X}}(\cdot, (a_i, x_i)) \right\|^2_{\mathcal{H}_{\mathcal{Y}}} + \lambda \| C \|^2_{HS}.$$

Let $\hat{C}_{Y|A,X} = \arg\min_C \hat{\mathcal{L}}^c(C)$. By kernelization,

$$\hat{\mu}_{Y|A,X}(a, x) \;=\; \hat{C}_{Y|A,X}\, k_{\mathcal{A}\mathcal{X}}(\cdot, (a, x)) \;=\; \sum_{i=1}^{n} \beta_i(a, x)\, k_{\mathcal{Y}}(\cdot, y_i),$$

$$\boldsymbol{\beta}(a, x) \;=\; \left(K_{AX,AX} + n\lambda I\right)^{-1} K_{AX,ax},$$

**Estimate $\hat{\mu}_\pi$.**

$$\text{Discrete } \mathcal{A}: \quad \hat{\mu}_\pi = \frac{1}{n}\sum_{i=1}^{n}\sum_{a\in\mathcal{A}} \pi(a \mid x_i)\, k_{\mathcal{A}\mathcal{X}}(\cdot, (a, x_i)).$$

$$\text{Continuous } \mathcal{A}: \quad \hat{\mu}_\pi = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{M}\sum_{m=1}^{M} k_{\mathcal{A}\mathcal{X}}(\cdot, (a_{im}, x_i)),\; a_{im} \sim \pi(\cdot \mid x_i).$$

# Plug-in Consistency and Rates (SRC & EVD)

**Proposition (Plug-in convergence rate in RKHS)**

*Under selection-on-observables, RKHS regularity, and standard source (smoothness $c \in (1, 3]$) and eigenvalue decay (complexity $b \in (0, 1]$) assumptions, choose*

$$\lambda \;=\; n^{-1/(c+1/b)}.$$

*Then, with high probability,*

$$\left\|\hat{\chi}_{\mathrm{pi}}(\pi) - \chi(\pi)\right\|_{\mathcal{H}_{\mathcal{Y}}} = \mathcal{O}_p\Big(r_C(n)\Big) = \mathcal{O}_p\Big(n^{-\frac{c-1}{2(c+1/b)}}\Big)\,.$$

**Interpretation.**

- Error is dominated by estimating $C_{Y|A,X}$ (operator regression); $\hat{\mu}_\pi$ attains $n^{-1/2}$.
- $b$ smaller $\Rightarrow$ slower spectral decay (harder problem); $c$ larger $\Rightarrow$ smoother operator (easier).
- Best rate $= n^{-1/4}$ when $c = 3$ (favorable smoothness).

*Teaser.* A **doubly robust** estimator of CPME achieves root-$n$ inference when nuisances are learned well (next section).

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# The Efficient Influence Function (EIF) for the CPME

**Goal.** Construct an estimator of $\chi(\pi)$ that is *first-order unbiased* and *efficient*. A naïve plug-in $\hat{\chi}_{\mathrm{pi}}$ inherits bias from estimated nuisances ($\hat{\mu}_{Y|A,X}$, $\hat{\pi}_0$); the EIF provides the optimal correction.

**EIF for the CPME (at the truth).**

$$\psi^{\pi}(y, a, x) = \frac{\pi(a \mid x)}{\pi_0(a \mid x)} \{k_{\mathcal{Y}}(\cdot, y) - \mu_{Y|A,X}(a, x)\} + \int \mu_{Y|A,X}(a', x)\pi(da' \mid x) - \chi(\pi).$$

**Intuition by components.**

- $\frac{\pi}{\pi_0}(k_{\mathcal{Y}}(\cdot, y) - \mu_{Y|A,X})$: *IPW residual* — reweights the observed feature $k_{\mathcal{Y}}(\cdot, y)$ to emulate sampling under $\pi$, removing regression bias.
- $\int \mu_{Y|A,X}(a', x)\pi(da' \mid x)$: *Regression correction* — restores the target conditional mean under $\pi$.
- $-\chi(\pi)$: *Centering term* ensuring $\mathbb{E}[\psi^{\pi}] = 0$, so that empirical averages yield unbiased corrections.

The EIF yields a **doubly robust** estimator, consistent if $\mu_{Y|A,X}$ or $\pi_0$ is well-specified.

# The Efficient Influence Function (EIF) for the CPME - II

Let $\widehat{P}$ denote the empirical law induced by the estimated nuisances $(\hat{\mu}_{Y|A,X},\, \hat{\pi}_0)$. The plug-in estimator $\widehat{\chi}_{\mathrm{pi}}(\pi) = \chi(\pi; \widehat{P})$ then inherits bias from these nuisance estimates.

By definition of EIF $\psi^\pi$ acts as the *gradient* of the functional $\chi(\pi; P)$ with respect to the data-generating law $P$. Since $\chi(\pi; P)$ is pathwise differentiable, it admits a first-order (linear) expansion around the true distribution $P_0$:

$$\chi(\pi; P) - \chi(\pi; P_0) = \mathbb{E}_{P_0}\left[\psi^\pi(Z)\, \frac{dP - dP_0}{dP_0}(Z)\right] + o(\|P - P_0\|).$$

Replacing $P_0$ by $\widehat{P}$ gives

$$\chi(\pi; P_0) - \chi(\pi; \widehat{P}) \;\approx\; \mathbb{E}_{\widehat{P}}[\psi^\pi(Z)],$$

meaning that the EIF $\psi^\pi$ captures how small estimation errors in the distribution $(\widehat{P} - P_0)$ translate into first-order changes in the parameter.

This expansion motivates the *one-step* or *doubly robust* estimator:

$$\widehat{\chi}_{\mathrm{dr}}(\pi) = \widehat{\chi}_{\mathrm{pi}}(\pi) + \tfrac{1}{n} \sum_{i=1}^{n} \widehat{\psi}^\pi(Z_i).$$

The EIF corrects the first order plug-in bias, yielding a **semiparametric efficient** estimator, achieving the minimum possible asymptotic variance.

# From Plug-in to One-Step: Double Robustness and Better Rates

**One-step DR estimator (using estimated nuisances):**

$$\hat{\chi}_{\mathrm{dr}}(\pi) = \hat{\chi}_{\mathrm{pi}}(\pi) + \frac{1}{n}\sum_{i=1}^{n}\left[\frac{\pi(a_i \mid x_i)}{\hat{\pi}_0(a_i \mid x_i)}\left\{k_{\mathcal{Y}}(\cdot, y_i) - \hat{\mu}_{Y|A,X}(a_i, x_i)\right\}\right.$$
$$\left. + \int \hat{\mu}_{Y|A,X}(a, x_i)\pi(da' \mid x_i) - \hat{\chi}_{\mathrm{pi}}(\pi)\right].$$

**Why it's better than plug-in:**

**Proposition (DR convergence rate in RKHS)**

- **Double robustness.** *Consistent if either $\hat{\pi}_0$ or $\hat{\mu}_{Y|A,X}$ is correct.*
- **Rate improvement.** *In $\mathcal{H}_{\mathcal{Y}}$,*

$$\left\|\hat{\chi}_{\mathrm{dr}}(\pi) - \chi(\pi)\right\| = \mathcal{O}_p\Big(n^{-1/2} + r_{\pi_0}(n)\, r_C(n)\Big).$$

  *If $r_{\pi_0}(n) \asymp r_C(n) \asymp n^{-1/4}$, the product is $n^{-1/2} \Rightarrow \sqrt{n}$-rate—improving over the plug-in's $n^{-1/4}$ ceiling.*
- **Bias cancellation.** *The IPW residual term removes first-order plug-in bias; the augmentation stabilizes variance and reduces model misspecification error.*
- **Efficiency.** *The EIF achieves the semiparametric efficiency bound (minimal asymptotic variance among regular estimators) under standard conditions.*

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# Application I: Testing with CPME, Distributional OPE

**Goal.** Compare counterfactual outcome distributions under two policies $\pi, \pi'$:

$$H_0 : \mathbb{E}_{P_\pi}\left[k_{\mathcal{Y}}(\cdot, Y)\right] = \mathbb{E}_{P_{\pi'}}\left[k_{\mathcal{Y}}(\cdot, Y)\right].$$

**EIF-based moment (difference).** Let

$$\mu_{Y|A,X}(a,x) := \mathbb{E}\big[k_{\mathcal{Y}}(\cdot, Y) \mid A = a, X = x\big], \qquad \beta_\pi(x) := \mathbb{E}_{a\sim\pi(\cdot|x)}[\mu_{Y|A,X}(a,x)].$$

Define the Hilbert-valued score

$$\varphi_{\pi,\pi'}(y,a,x) = \left\{\frac{\pi(a|x)}{\pi_0(a|x)} - \frac{\pi'(a|x)}{\pi_0(a|x)}\right\}\Big\{k_{\mathcal{Y}}(\cdot, y) - \mu_{Y|A,X}(a,x)\Big\} + \beta_\pi(x) - \beta_{\pi'}(x).$$

Then

$$H_0 \quad \Longleftrightarrow \quad \mathbb{E}[\varphi_{\pi,\pi'}(Y,A,X)] = 0.$$

**Why EIF?**

- *Doubly robust:* unbiased if either $\pi_0$ or $\mu_{Y|A,X}$ is correctly specified.
- *Orthogonal moment:* locally insensitive to nuisance errors, yielding higher power with learned nuisances.

**Motivation.** Before deployment, test—using logged data—whether a new policy $\pi$ significantly alters the *entire* outcome distribution compared to the baseline $\pi_0$.

# Application I: DR-KPT - cross U-statistic and decision rule

**Split & estimate.** Split the sample: first $m = \lfloor n/2 \rfloor$, last $n - m$. Fit $\hat{\pi}_0, \hat{\mu}_{Y|A,X}$ on each split separately and compute $\hat{\varphi}$ on the first split, $\tilde{\varphi}$ on the second.
**Cross U statistic.** For $i \leq m$,

$$f^{\dagger}_{\pi,\pi'}(Y_i, A_i, X_i) = \frac{1}{n-m} \sum_{j=m+1}^{n} \left\langle \hat{\varphi}_{\pi,\pi'}(Y_i, A_i, X_i),\, \tilde{\varphi}_{\pi,\pi'}(Y_j, A_j, X_j) \right\rangle_{\mathcal{H}_{\mathcal{Y}}}.$$

Let

$$\bar{f}^{\dagger} = \frac{1}{m} \sum_{i=1}^{m} f^{\dagger}_{\pi,\pi'}(Y_i, A_i, X_i), \qquad S^{\dagger} = \sqrt{\frac{1}{m} \sum_{i=1}^{m} \left(f^{\dagger}_{\pi,\pi'} - \bar{f}^{\dagger}\right)^2},$$

$$\boxed{T^{\dagger}_{\pi,\pi'} = \frac{\sqrt{m}\, \bar{f}^{\dagger}}{S^{\dagger}}}.$$

**Proposition (Asymptotic normality & decision.)**

*Under regularity and* $r_{\pi_0}(n)\, r_C(n) = \mathcal{O}(n^{-1/2})$, $T^{\dagger}_{\pi,\pi'} \Rightarrow \mathcal{N}(0,1)$. *Compute* $p = 1 - \Phi(T^{\dagger}_{\pi,\pi'})$; *reject* $H_0$ *at level* $\alpha$ *if* $p \leq \alpha$.

**Notes.** Cross U handles MMD degeneracy; $m = \lfloor n/2 \rfloor$ maximizes power; EIF-based (DR) estimation boosts rate and sensitivity versus pure plug-in or IPW.

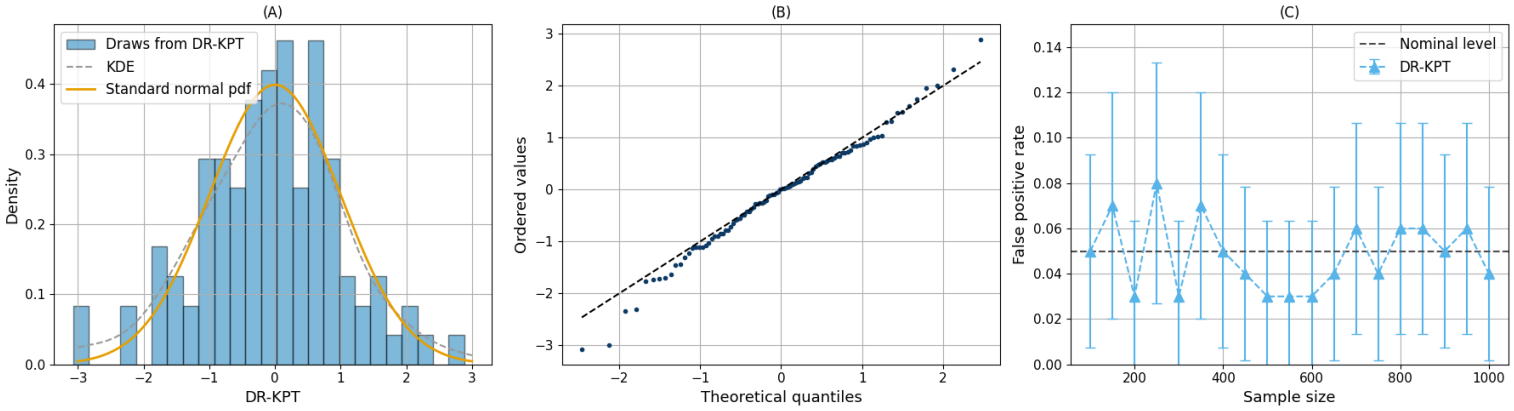
# Application I: DR-KPT - experimental setup & calibration

**Data.** i.i.d. $(X_i, A_i, Y_i)_{i=1}^n$; continuous actions; unknown nuisances.
**Nuisance learning.** $\hat{\pi}_0(\cdot \mid x)$: linear regression; $\hat{\mu}_{Y|A,X}$: conditional mean embeddings.
**Level.** All tests at $\alpha = 0.05$. 100 simulations.

**Scenarios.**

- *I (Null)*: $\pi = \pi' \Rightarrow P_{Y(\pi)} = P_{Y(\pi')}$.
- *II (Mean shift)*: $P_{Y(\pi)}, P_{Y(\pi')}$ differ in means.
- *III (Mixture)*: $P_{Y(\pi')}$ is a mixture; mean unchanged compared to $P_{Y(\pi)} \Rightarrow$ bimodal shift.
- *IV (Shifted mixture)*: mixture + mean shift.

**Calibration under the null (Scenario I).** DR-KPT statistic $T^{\dagger}_{\pi,\pi'}$ is approximately $\mathcal{N}(0,1)$ and the empirical type-I error matches 0.05.



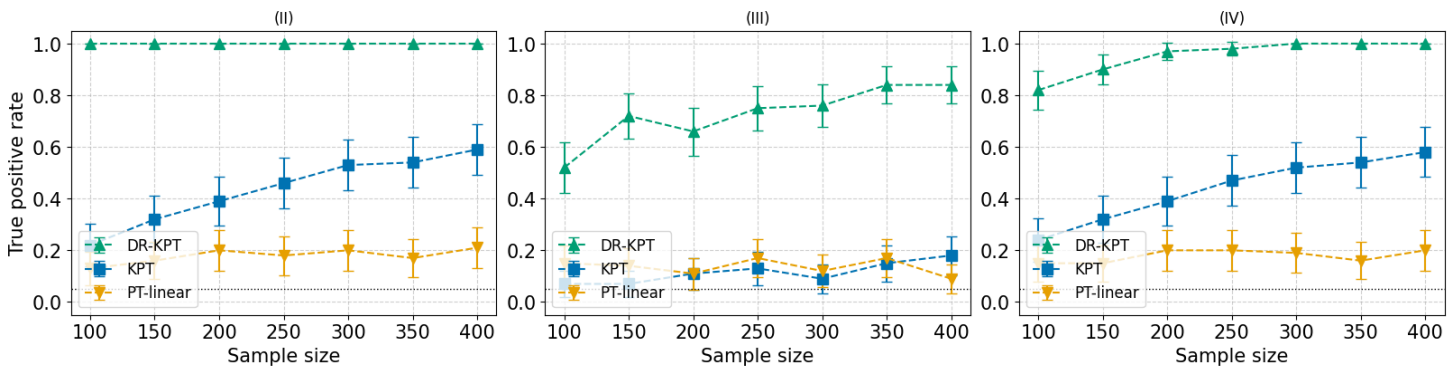
# DR-KPT vs. baselines: power under Scenarios II–IV

**Baselines.**

- **KPT** (adapted from KTE, *estimated* propensities).
- **PT-linear**: linear kernel (tests only mean differences).

**Result.** DR-KPT shows higher true-positive rates in all alternatives:

- Detects pure mean shifts (Scenario II) and non-mean shifts (bimodality, Scen. III) as well as combined shifts (Scen. IV).
- KPT / PT-linear often lose power due to propensity estimation error or insensitivity to higher moments.
- EIF-based, doubly robust construction $\Rightarrow$ better calibration + power with learned nuisances; cross-U handles MMD degeneracy; computation is efficient.



# Application I: Runtime — DR-KPT is *orders of magnitude* faster

**Summary (Scenarios I–IV). DR-KPT** achieves the best computational efficiency—typically $\mathbf{10^2}\times$ faster— thanks to its *closed-form* test statistic, avoiding costly resampling or kernel-matrix permutations.

**Why:** KPT relies on permutation-based MMD with repeated kernel recomputation, while PT-linear still requires multiple reweighting steps. In contrast, DR-KPT computes a single analytic statistic with no resampling.

**Table:** Average runtime (s) for Scenario I (mean $\pm$ std over 100 runs).

| Sample size | KPT | PT-linear | DR-KPT |
|---|---|---|---|
| **100** | 0.495 $\pm$ 0.070 | 0.592 $\pm$ 0.061 | **0.004 $\pm$ 0.005** |
| **150** | 0.740 $\pm$ 0.039 | 0.774 $\pm$ 0.038 | **0.007 $\pm$ 0.004** |
| **200** | 1.134 $\pm$ 0.081 | 1.060 $\pm$ 0.051 | **0.010 $\pm$ 0.009** |
| **250** | 1.623 $\pm$ 0.075 | 1.553 $\pm$ 0.076 | **0.008 $\pm$ 0.002** |
| **300** | 2.257 $\pm$ 0.074 | 2.373 $\pm$ 0.202 | **0.013 $\pm$ 0.007** |
| **350** | 3.204 $\pm$ 0.118 | 3.384 $\pm$ 0.160 | **0.025 $\pm$ 0.023** |
| **400** | 4.180 $\pm$ 0.136 | 4.358 $\pm$ 0.251 | **0.019 $\pm$ 0.007** |

*DR-KPT's closed-form expression removes the need for kernel recomputation or randomization, making it up to two orders of magnitude faster than classical kernel tests.*

# Application II - Sampling from the counterfactual distribution

**Goal.** Use the estimated embedding $\hat{\chi}(\pi) \in \mathcal{H}_{\mathcal{Y}}$ to generate pseudo-samples $\tilde{y}_{1:m}$ whose empirical law approximates the outcome distribution under $\pi$.

**Deterministic kernel herding (from $\hat{\chi}(\pi)$).**

$$\tilde{y}_1 := \arg\max_{y \in \mathcal{Y}} \hat{\chi}(\pi)(y), \qquad \tilde{y}_t := \arg\max_{y \in \mathcal{Y}} \left[\hat{\chi}(\pi)(y) - \frac{1}{t}\sum_{\ell=1}^{t-1} k_{\mathcal{Y}}(\tilde{y}_\ell, y)\right], \; t = 2, \ldots, m.$$

**Proposition (Guarantees (MMD & weak convergence))**

*Under mild regularity (locally compact $\mathcal{Y}$, $\mathcal{H}_{\mathcal{Y}} \subset \mathcal{C}_0$):*

$$\mathrm{MMD}\left(\tilde{P}^m_{Y,pi}, \nu(\pi)\right) = \mathcal{O}_p\left(r_C(n,b,c) + m^{-1/2}\right),$$

$$\mathrm{MMD}\left(\tilde{P}^m_{Y,dr}, \nu(\pi)\right) = \mathcal{O}_p\left(n^{-1/2} + r_{\pi_0}(n) r_C(n,b,c) + m^{-1/2}\right),$$

$\Rightarrow \tilde{y}_{pi,j} \rightsquigarrow \nu(\pi)$ *and* $\tilde{y}_{dr,j} \rightsquigarrow \nu(\pi)$.

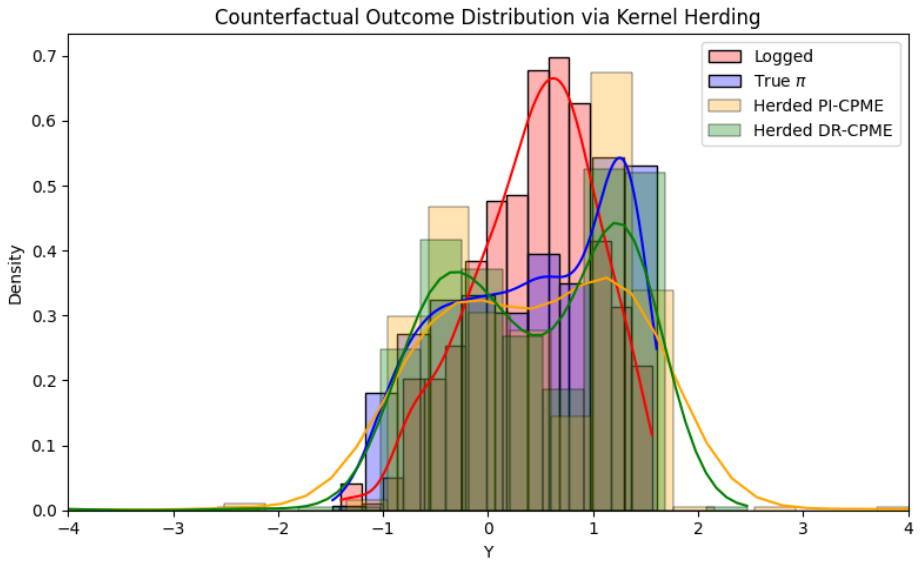
$m$ trades compute for quality; DR embedding improves sample quality via the faster $n^{-1/2}$-driven term.

**Motivation.** In simulation studies, we can arbitrarily sample from any given policy to assess the behavior of our test under controlled conditions—allowing us to validate its sensitivity to distributional shifts.

# Application II: Sampling via kernel herding - setup & qualitative result

**Goal.** Generate counterfactual samples from $\hat{\chi}(\pi) \in \mathcal{H}_{\mathcal{Y}}$ using kernel herding; compare **Plug-in** vs. **DR** CPME.
**Scenarios.** Logging policy $\pi_0 \in \{\text{uniform, logistic}\}$; outcome $g \in \{\text{quadratic, sinusoidal (nonlinear)}\}$.

- Herd from $\hat{\chi}_{\mathrm{pi}}(\pi)$ and $\hat{\chi}_{\mathrm{dr}}(\pi)$ (Alg. sampling).
- Compare herded empirical law $\tilde{P}_Y^m$ to oracle $\nu(\pi)$ (synthetic ground truth).
- Metrics: Wasserstein distance and MMD (Gaussian kernel).




Logistic $\pi_0$, nonlinear outcome: logged vs. oracle vs. Plug-in/DR herded.

# Application II: Sampling fidelity - Wasserstein & MMD (herded vs. oracle)

**Wasserstein distance**

| Method | log–nonlin | log–quad | uni–nonlin | uni–quad |
|---|---|---|---|---|
| Plug-in | 1.29e-01 ± 2.6e-01 | 1.41e-01 ± 4.9e-02 | 9.08e-02 ± 3.7e-01 | 6.78e-02 ± 1.9e-02 |
| DR | **8.60e-02** ± 2.2e-02 | 1.36e-01 ± 3.9e-02 | **5.00e-02** ± 1.5e-02 | **6.63e-02** ± 1.6e-02 |

**MMD (Gaussian kernel)**

| Method | log–nonlin | log–quad | uni–nonlin | uni–quad |
|---|---|---|---|---|
| Plug-in | 1.11e-03 ± 5.9e-03 | 9.85e-04 ± 6.0e-04 | 1.92e-04 ± 1.2e-03 | 3.31e-04 ± 2.5e-04 |
| DR | **4.38e-04** ± 3.6e-04 | 9.80e-04 ± 6.0e-04 | **6.49e-05** ± 4.4e-05 | 3.51e-04 ± 2.5e-04 |

**Takeaways.**

- DR herding matches oracle better (lower Wasserstein/MMD), especially under *nonlinear* outcomes and *logistic* logging where misspecification is hardest.
- Consistent with theory: DR CPME improves the embedding ($n^{-1/2}$-driven term), yielding higher-fidelity samples from $\nu(\pi)$.

# Discussion

**Summary.** We introduced the **Counterfactual Policy Mean Embedding (CPME)** (Zenati et al., 2025) — a general kernel-based framework for *distributional off-policy evaluation*. It unifies plug-in and **EIF-based doubly robust** estimation, enables the **DR-KPT** test (a closed-form, non-permutation kernel test), and supports deterministic counterfactual sampling via kernel herding.

**Limitations.**

- Standard identification assumptions: consistency, exchangeability, and positivity.
- RKHS regularity and kernel choice; computational scalability of kernel operations ($O(n^2)$).

**Future work.**

- *Distributional policy optimization*: learn policies by optimizing CPMEs under risk or safety constraints.
- *Robustness*: develop sensitivity analyses, proximal or instrumental-variable extensions, and partial identification bounds for $\chi(\pi)$.

*Ongoing direction:* extending kernel treatment effects to **adaptively collected data**. New preprint and code: `https://github.com/houssamzenati/adaptive-KTE`

# References


 Muandet, Krikamol et al. (2021). "Counterfactual mean embeddings". In: *Journal of Machine Learning Research* 22.162, pp. 1–71.

 Zenati, Houssam et al. (2025). "Doubly-Robust Estimation of Counterfactual Policy Mean Embeddings". In: *Neural Information Processing Systems*.

# The Efficient Influence Function (EIF) for the CPME – I

**Pathwise view.** Consider a regular submodel $\{P_\varepsilon : \varepsilon \in \mathbb{R}\}$ through the true law $P_0$, with density

$$p_\varepsilon(z) = p_0(z)\{1 + \varepsilon s(z)\} + o(\varepsilon),$$

where $s(Z) \in L^2(P_0)$ is the **score function**. It encodes a valid infinitesimal perturbation of $P_0$: the directions in which the data law can move smoothly.

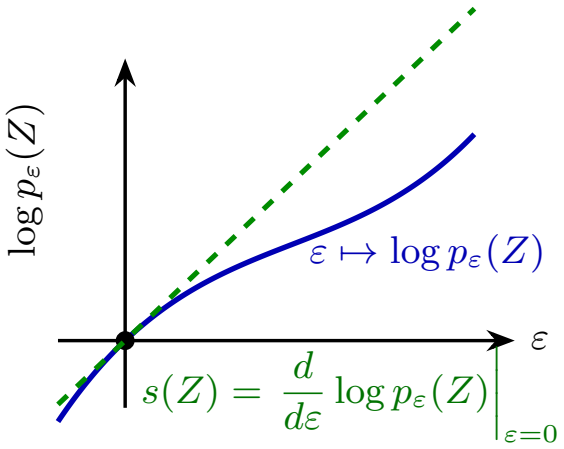
The parameter path $\varepsilon \mapsto \chi_\varepsilon(\pi) \in \mathcal{H}_\mathcal{Y}$ is **pathwise differentiable** if for every score $s$,

$$D_s\chi(\pi) := \left.\frac{d}{d\varepsilon}\chi_\varepsilon(\pi)\right|_0 \text{ exists in } \mathcal{H}_\mathcal{Y}.$$

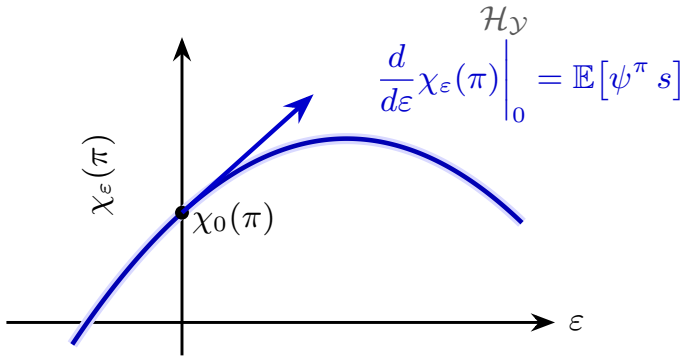
By the Riesz representation theorem, there exists a unique $\psi^\pi(Y, A, X) \in L^2(P_0; \mathcal{H}_\mathcal{Y})$ such that

$$D_s\chi(\pi) = \mathbb{E}_{P_0}[\psi^\pi s],$$

defining the **efficient influence function (EIF)** as the Riesz representer of the pathwise derivative.




**Left:** $s(Z) = \partial_\varepsilon \log p_\varepsilon(Z)|_0$ is the slope of the log-density; it specifies the direction of infinitesimal change in $P_\varepsilon$.




**Right:** The parameter path $\varepsilon \mapsto \chi_\varepsilon(\pi)$ in $\mathcal{H}_\mathcal{Y}$; its tangent at 0 is $\mathbb{E}[\psi^\pi s]$, defining the EIF as the functional "gradient" of $\chi(\pi)$ with respect to the data law.

# The Efficient Influence Function (EIF) for the CPME – II

**Influence functions.** An *influence function* $h(Z) \in L^2(P_0; \mathcal{H}_\mathcal{Y})$ characterizes how a parameter $\chi(P)$ reacts to infinitesimal perturbations of the data law:

$$\frac{d}{d\varepsilon}\chi(P_\varepsilon)\Big|_0 = \mathbb{E}_{P_0}[h(Z)s(Z)], \qquad \mathbb{E}_{P_0}[h(Z)] = 0.$$

It quantifies how each observation $Z$ contributes to the change in $\chi(P)$ when $P$ is locally perturbed along the *score direction* $s(Z)$.

**Why the tangent space.** Let $\mathcal{T} \subseteq L^2(P_0)$ denote the **tangent space**—the closure of all valid score functions. It represents the set of directions in which the distribution $P_0$ can vary smoothly within the model. Projecting onto $\mathcal{T}$ ensures that we only consider directions that correspond to *real, estimable* perturbations of the data law.

**Geometric interpretation.** The **EIF** $\psi^\pi \in L^2(P_0; \mathcal{H}_\mathcal{Y})$ is the *unique element* whose inner product with any score reproduces the pathwise derivative:

$$D_s\chi(\pi) = \mathbb{E}_{P_0}[\psi^\pi s], \quad \forall s \in \mathcal{T}.$$

Equivalently, $\psi^\pi$ is the *orthogonal projection* of any regular influence function $h$ onto the tangent space $\mathcal{T}$—it isolates the part of $h$ that corresponds to meaningful, model-consistent variation in $P$.

# The Efficient Influence Function (EIF) for the CPME – III

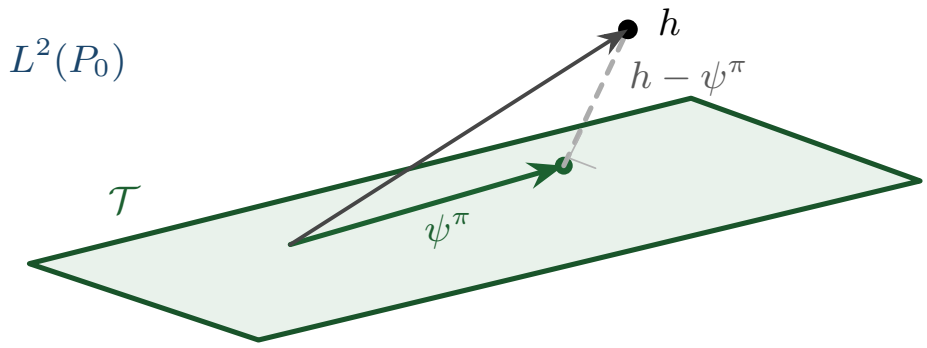



**Figure:** The EIF $\psi^\pi$ is the orthogonal projection of any regular influence function $h$ onto the tangent space $\mathcal{T} \subset L^2(P_0)$. The residual $h - \psi^\pi$ is orthogonal to $\mathcal{T}$, meaning that it corresponds to directions that cannot change $\chi(P)$.

**Why it minimizes variance.** Since $h - \psi^\pi$ is orthogonal to all valid scores:

$$\mathbb{E}_{P_0}[(h - \psi^\pi)s] = 0, \quad \forall s \in \mathcal{T},$$

the EIF $\psi^\pi$ is the projection of $h$ onto $\mathcal{T}$. By the Pythagorean theorem in $L^2(P_0)$,

$$\|h\|^2_{L^2(P_0)} = \|\psi^\pi\|^2_{L^2(P_0)} + \|h - \psi^\pi\|^2_{L^2(P_0)} \Rightarrow \|h\|^2_{L^2(P_0)} \geq \|\psi^\pi\|^2_{L^2(P_0)}.$$

Thus, the EIF yields the **smallest possible asymptotic variance**: it is the most efficient (shortest) direction in $L^2(P_0)$ that still preserves the correct local behavior of $\chi(\pi)$ under valid perturbations of $P$.